

M.B.

B.E.

F.D.

(2)

Applies to system of

Identical, distinguishable Particles

Identical Indistinguishable Particles, do not obey Pauli-exclusion principle

Identical, Indistinguishable obey Pauli's principle

Category of Particle

Classical (Gas molecules)

Bosons (Photons etc)

Fermions (Protons, Neutrons, Electrons)

Example.

molecules of gas

Photons in a cavity, lig. He at low Temp.

Free Electrons in metals, Protons, Neutrons

Properties of Particles

wave function do not overlap at any spin

Spin 0, ±1, wave function are symmetric to interchanging of Particles

Spin ±1/2, wave function are antisymmetric to interchange of Particles

$$\psi(2,1) = \psi(1,2)$$

$$\psi(2,1) = -\psi(1,2)$$

Distribution

$$n_i = \frac{g_i}{e^{\alpha + \beta \epsilon_i}}$$

$$n_i = \frac{g_i}{e^{\alpha + \beta \epsilon_i} - 1}$$

$$n_i = \frac{g_i}{e^{\alpha + \beta \epsilon_i} + 1}$$

Properties of Distribution function of Particles

No limit of number of Particles per state

No limit to Number of Particles per State

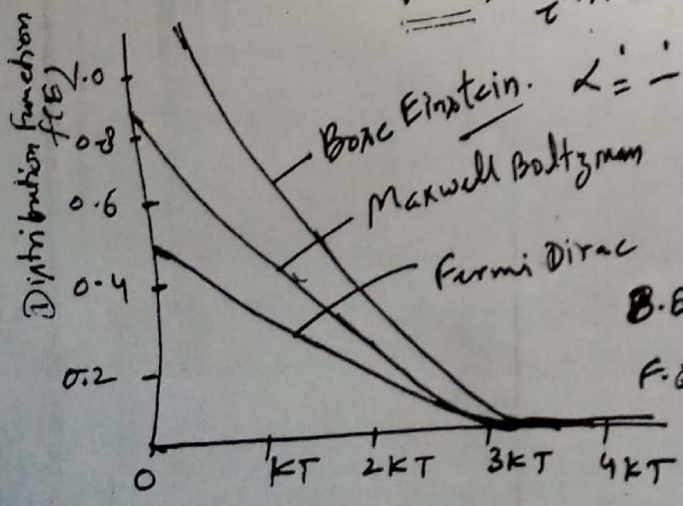
Never more than 1-particle per state.

Distribution function

$$f_{MB} = \frac{1}{e^{\alpha + \beta \epsilon_i}}$$

$$f_{BE} = \frac{1}{e^{\alpha + \beta \epsilon_i} - 1}$$

$$f_{FD} = \frac{1}{e^{\alpha + \beta \epsilon_i} + 1}$$



Bose Einstein

Maxwell Boltzmann

Fermi Dirac

$$\alpha = -\frac{\mu}{KT}$$

$$\beta = \frac{1}{KT}$$

K = Boltzmann const

μ = Chemical Pot. per mole

(Rate of change of free energy per mole at const. Vol. + const. Temp (Fermi Energy))

M.B.

P	q
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q	p
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q	q
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p	q
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B.E.

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F.D.

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$$n(p) dp = \frac{g(p) dp}{\exp(\alpha) - \exp(\beta \epsilon)}$$

$$\epsilon = \frac{p^2}{2m}$$

$$n(p) dp = \frac{g(p) dp}{\exp(\alpha) - \exp\left(-\frac{\beta p^2}{2m}\right)}$$

Here $n(p) dp$ represents the number of molecules having momentum b/w p and $p+dp$ and $g(p) dp$ is the number of cells in phase space corresponding to momentum range dp .

Volume of cell in phase space

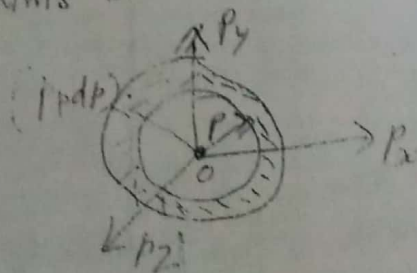
$$dV = dx dy dz dp_x dp_y dp_z = h^3$$

$$g(p) dp = \frac{\iiint dx dy dz \iiint dp_x dp_y dp_z}{h^3}$$

$$g(p) dp = \frac{\iiint dx dy dz \iiint dp_x dp_y dp_z}{h^3}$$

$$g(p) dp = \frac{\iiint dp_x dp_y dp_z}{h^3}$$

Since $g(p) dp$ is the number of cells in the phase space in momentum range from p to $p+dp$, we can find the value of $\iiint dp_x dp_y dp_z$ by drawing spheres of radii p and $p+dp$ with their centers at the origin of momentum space. In this condition, the volume between the two spheres represents all the molecules having momentums between p and $p+dp$.



Thus we have

$\iiint dp_x dp_y dp_z = \text{Vol. of spherical shell enclosed b/w the two spheres.}$

$\iiint dp_x dp_y dp_z = (\text{Surface area of the sphere of radius } p) \times (\text{thickness of shell})$

$$\iiint dp_x dp_y dp_z = (4\pi p^2) dp$$

$$= 4\pi p^2 dp$$

$$g(p) dp = \frac{4\pi V p^2 dp}{h^3}$$

$$n(p) dp = \frac{4\pi V p^2}{h^3} \exp\left(-\frac{\beta p^2}{2m}\right) dp$$

$$E = \frac{1}{2} m v^2$$

$$p = m v$$

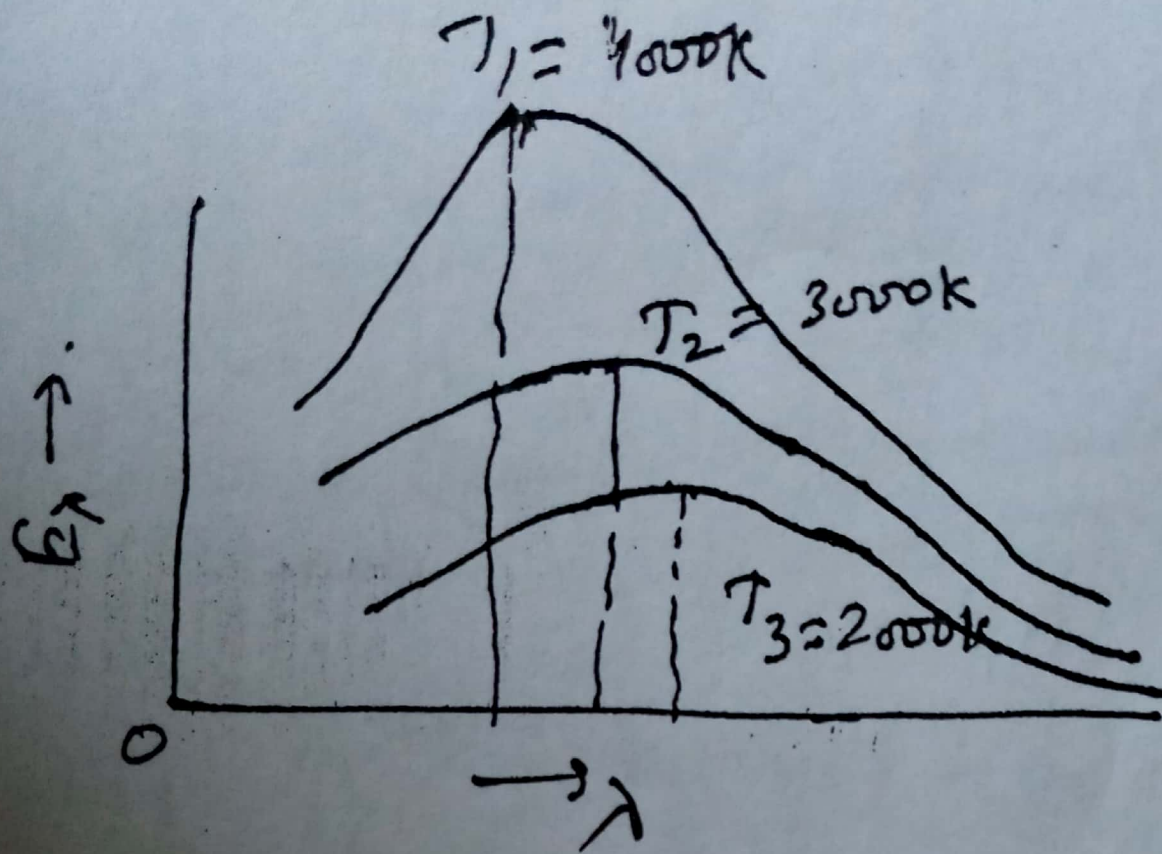
$$p^2 = m^2 v^2$$

$$\frac{p^2}{m} = m v^2$$

$$\Delta x \cdot \Delta p_x \geq \frac{h}{2\pi}$$

$$\Delta x \cdot \Delta p_x \geq \frac{h}{2\pi}$$

$$\Delta x \cdot \Delta p_x \geq \frac{h}{2\pi}$$



Black Body Radiation:

Rayleigh Jeans, Planck's & Wein

ν $\rightarrow$ No. of molecules having momentum b/w p & $p+dp$

dp $\rightarrow$ No. of cells in Phase Space corresponding to momentum range dp

V $\rightarrow$ Volume of each cell in Phase space (Eigen state)

$$dx dy dz dp_x dp_y dp_z = h^3$$

$$N dp = \frac{\iiint dx dy dz \iiint dp_x dp_y dp_z}{h^3}$$

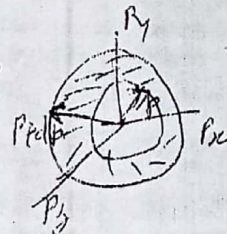
$$= \frac{\iiint dx dy dz \iiint dp_x dp_y dp_z}{h^3}$$

$$= \frac{V \iiint dp_x dp_y dp_z}{h^3}$$

$\iiint dp_x dp_y dp_z =$ Vol. of Sp. Shell enclosed b/w the two spheres.

= Surface area of the sphere of radius p x thickness of shell

$$\iiint dp_x dp_y dp_z = 4\pi p^2 dp$$



$$N dp = V \cdot 4\pi p^2 dp$$

Photon (freq ν) $h\nu$

$$mc = \frac{h\nu}{c} \text{ or } dp = \frac{h\nu}{c}$$

$$g(\nu) d\nu = \frac{V \cdot 4\pi (\frac{h\nu}{c})^2 \cdot \frac{h d\nu}{c}}{h^3}$$

$$g(\nu) d\nu = \frac{V \cdot 4\pi \cdot h^3 \nu^2 \cdot h d\nu}{h^3 \times c^3}$$

$$= \frac{16\pi V \nu^2 d\nu}{c^3}$$

+ High Temp. ΔT is large $\rightarrow$ $\frac{h\nu}{kT} \rightarrow 0$

Due to two modes of Propagations of Photons (x, p). The total No. of eigen state in the freq. range ν to $(\nu+d\nu)$

$$g(\nu) d\nu = \frac{8\pi V \nu^2 d\nu}{c^3} \quad (3)$$

The Energy per unit volume in the freq. range ν to $(\nu+d\nu)$ is

$$E(\nu) d\nu = \epsilon \cdot g(\nu) d\nu$$

$$= \frac{KT \cdot g(\nu) d\nu}{V} = \frac{8\pi V^2 KT d\nu}{c^3}$$

(Rayleigh Jeans Law)

B-E (Bose-Einstein)

$$n_i = \frac{g_i}{e^{\alpha + \beta \epsilon_i} - 1}$$

In freq. range ν to $(\nu+d\nu)$

$$n(\nu) d\nu = \frac{g(\nu) d\nu}{e^{\alpha + \beta \epsilon} - 1}$$

In this case $\alpha = 0, \beta = \frac{1}{KT}, \epsilon = h\nu$

$$n(\nu) d\nu = \frac{g(\nu) d\nu}{e^{\frac{h\nu}{KT}} - 1}$$

$$n(\nu) d\nu = \frac{8\pi V \nu^2 d\nu}{c^3} \cdot \frac{1}{e^{\frac{h\nu}{KT}} - 1} \quad \left| \frac{g(\nu) d\nu}{c^3} = \frac{8\pi V \nu^2 d\nu}{c^3} \right|$$

$$\frac{n(\nu) d\nu}{V} = \frac{8\pi \nu^2 d\nu}{c^3} \cdot \frac{1}{e^{\frac{h\nu}{KT}} - 1}$$

Energy of one photon

$$E(\nu) d\nu = \frac{n(\nu) d\nu}{V} \times h\nu$$

$$= \frac{8\pi \nu^2 d\nu}{c^3} \cdot \frac{h\nu}{e^{\frac{h\nu}{KT}} - 1}$$

(Planck's Radiation Law)

$$E(\nu) d\nu = \frac{8\pi \nu^2 d\nu}{c^3} \cdot \frac{h\nu}{[e^{\frac{h\nu}{KT}} - 1]} = \frac{8\pi V^2 KT d\nu}{c^3}$$

(Rayleigh-Jeans Law)

$$\bar{E}_\gamma = \int_0^\infty E(\gamma) d\gamma = \int_0^\infty \frac{8\pi h \nu^3 d\nu}{c^3 (e^{h\nu/KT} - 1)} = \frac{8\pi h^4}{15 c^3 K^4} T^4$$

$$E(\gamma) d\gamma = \frac{8\pi h \nu^3 d\nu}{c^3 (e^{h\nu/KT} - 1)}$$

Expressing in the form of wavelength.
(At ~~low~~ temp.)
 $\nu = \frac{c}{\lambda} \quad \text{and} \quad d\nu = -\frac{c d\lambda}{\lambda^2}$

$$E(\lambda) d\lambda = \frac{8\pi h \left(\frac{c}{\lambda}\right)^3 \lambda \cdot \frac{c d\lambda}{\lambda^2}}{c^3 (e^{hc/\lambda KT} - 1)}$$

$$E_\lambda d\lambda = - \frac{8\pi h c}{\lambda^5} \frac{d\lambda}{e^{hc/\lambda KT} - 1}$$

$$E_\lambda = \frac{8\pi h c}{\lambda^5} \frac{1}{e^{hc/\lambda KT} - 1} \quad \text{Wein Displ. Law}$$

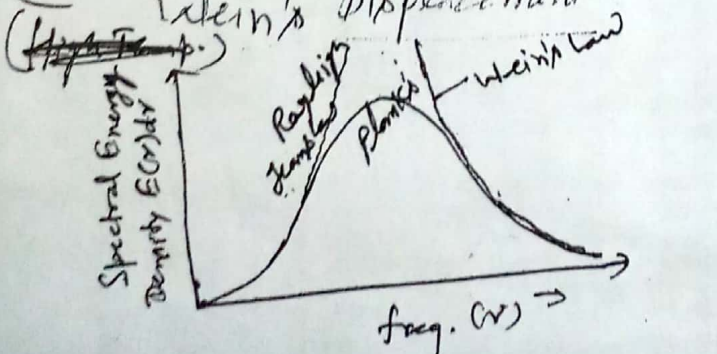
For $\lambda_{\max} \rightarrow \frac{dE_\lambda}{d\lambda} = 0$

$$\frac{hc}{\lambda_{\max} KT} = 4.965 \text{ (const.)}$$

$$\lambda_{\max} T = \frac{hc}{4.965 K} = 2.898 \times 10^{-3} \text{ m.K}$$

$$\lambda_{\max} T = \text{const.} \Rightarrow \lambda_m \propto \frac{1}{T}$$

(Low Temp.)
(High Temp.)



Stefan Boltzmann Law
Fermi Dirac - $F(E) = \frac{1}{e^{(E-E_F)/kT} + 1}$

$$g(p) dp = \frac{8\pi V p^2 dp}{h^3}$$

$$E = \frac{p^2}{2m} \Rightarrow p = \sqrt{2mE}$$

$$dp = \frac{\sqrt{2m}}{2\sqrt{E}} dE \Rightarrow dp = \frac{1}{2} (2m/E)^{1/2} dE$$

$$g(E) dE = \frac{4\pi V \times \sqrt{2mE} \cdot \frac{1}{2} (2m/E)^{1/2} dE}{h^3}$$

$$= \frac{8\pi V \times \sqrt{2m} \cdot \sqrt{E} \cdot \frac{1}{2} (2m)^{1/2} dE}{h^3}$$

$$g(E) dE = \frac{8\sqrt{2} \pi V m^{3/2}}{h^3} E^{1/2} dE$$

However, the number of Electrons that can have the same Energy.

$$N = \int_0^{E_F} g(E) dE$$

$$N = \frac{8\sqrt{2} \pi V m^{3/2}}{h^3} \times \frac{2}{3} E_F^{3/2}$$

$$N = \frac{16\sqrt{2} \pi V m^{3/2}}{3 h^3} E_F^{3/2}$$

$$E_F = \left[\frac{3 N h^3}{16\sqrt{2} \pi V m^{3/2}} \right]^{2/3}$$

$$E_F = \frac{h^2}{2m} \left(\frac{3 N}{8\pi V} \right)^{2/3}$$

Fermi Energy

$$\frac{dE}{d\lambda} = \frac{8\pi h c}{\lambda^5} \cdot \left[-\frac{1}{(e^{hc/\lambda kT} - 1)^2} \cdot e^{hc/\lambda kT} \cdot \left(-\frac{hc}{\lambda^2 kT}\right) \right]$$

$$- \frac{8\pi h c \times 5}{\lambda^6} \left[\frac{1}{e^{hc/\lambda kT} - 1} \right] = 0$$

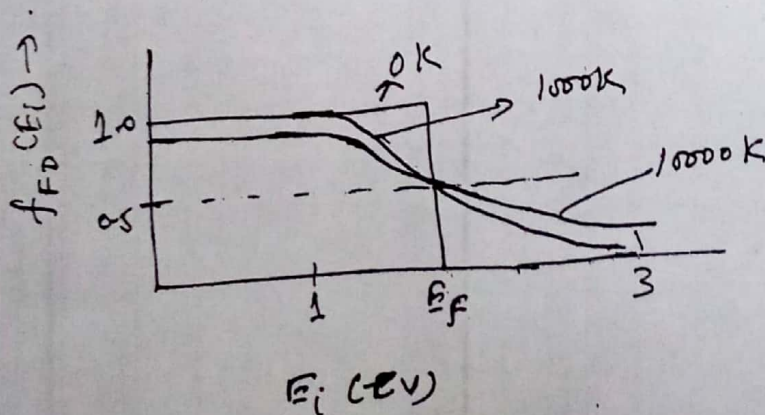
$$\Rightarrow \frac{8\pi h c}{\lambda^5} \left[\frac{hc}{\lambda^2 kT} \frac{e^{hc/\lambda kT}}{(e^{hc/\lambda kT} - 1)^2} \right] = \frac{8\pi h c \times 5}{\lambda^6} \frac{1}{e^{hc/\lambda kT} - 1}$$

$$\frac{hc}{\lambda kT} \cdot \frac{e^{hc/\lambda kT}}{e^{hc/\lambda kT} - 1} = 5$$

$$\frac{hc}{\lambda kT} = \frac{(e^{hc/\lambda kT} - 1)}{e^{hc/\lambda kT}} \cdot 5$$

$$= \left[1 - \frac{1}{e^{hc/\lambda kT}} \right] 5$$

$$= 5 - 5 e^{-\frac{hc}{\lambda kT}}$$



F.D. Distribution Curve